

Silence in the Spectrum: Unmasking Speech From Noise

Junze Xie | Ming Chen | Zifan Xu | Yimin Chen | Wanting Luo

Signals, Systems & Probability

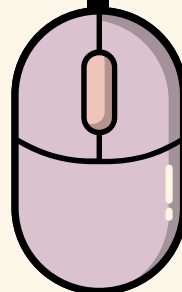




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01

Background & Problems

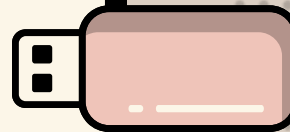
Voice Denoising in
Physical Environment



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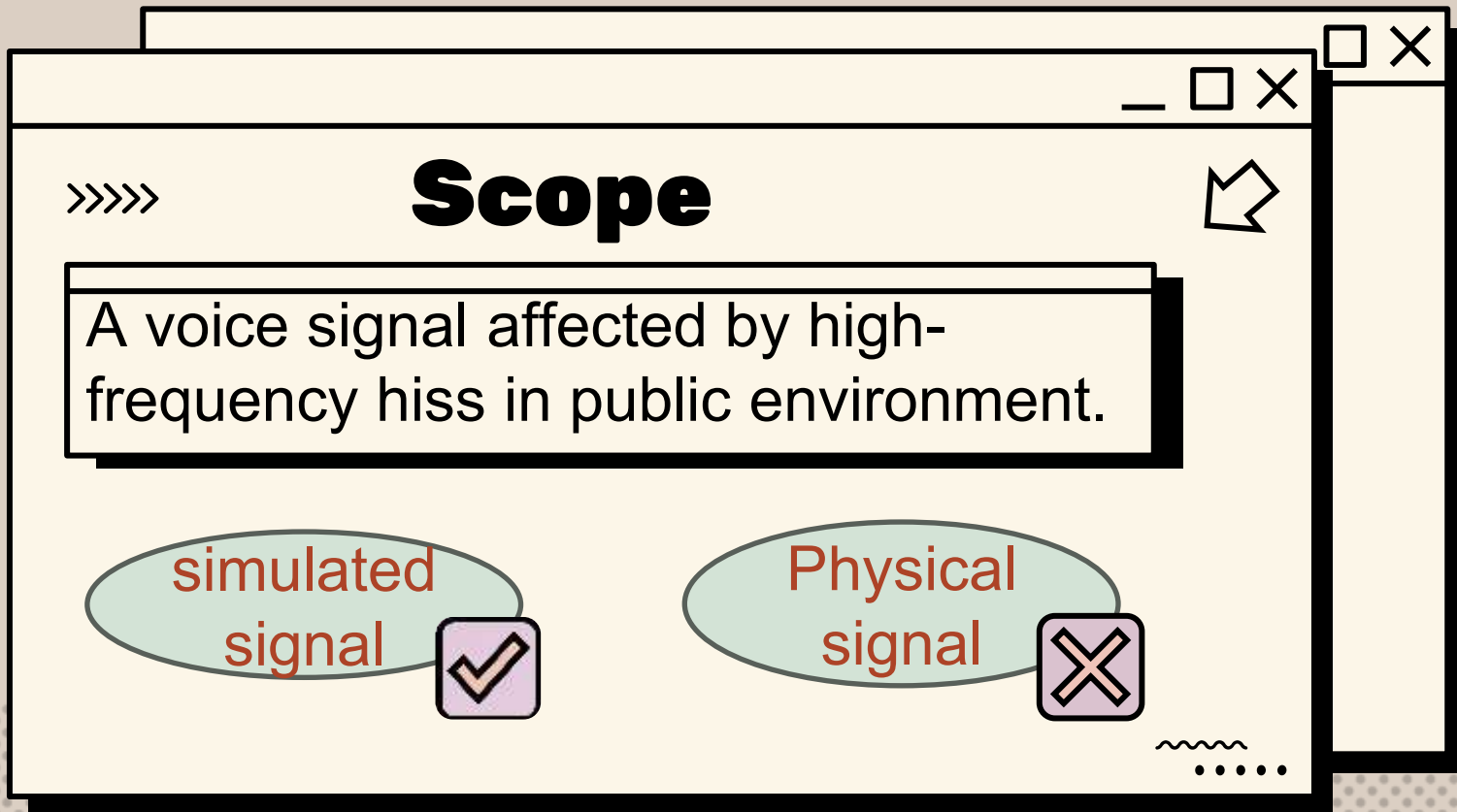


# Scenario

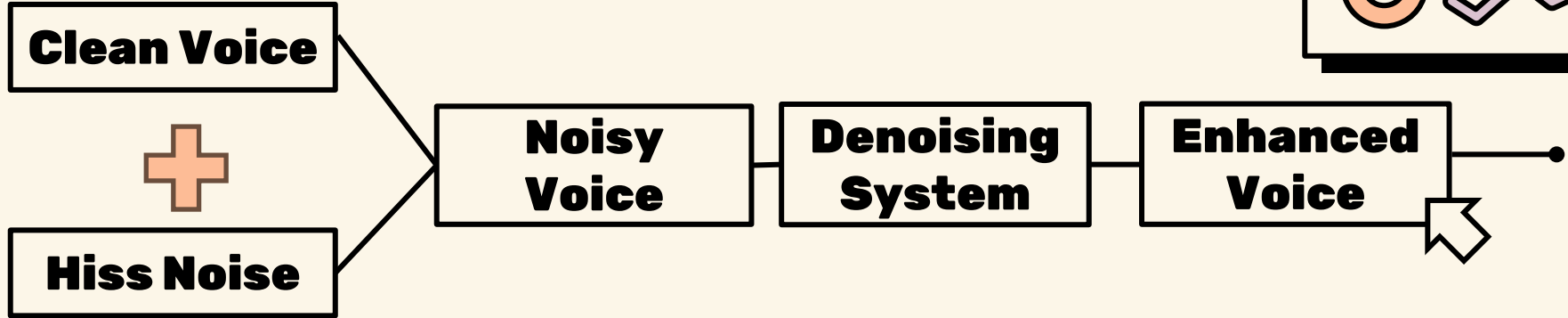
In a public setting  
Such as a train station  
A phone call came  
through

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# ..... **Problem Statement**



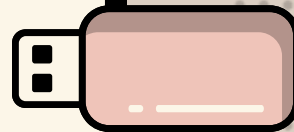
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**Goal: remove the noise from background
and preserve the voice from user.**

02

Signal Model & Baseline Filter

Fundamentals of Speech Denoising



Signal Model

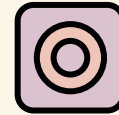
$$X_{noise}[n] = X_{clean}[n] + N_{hiss}[n]$$



– The clean and hiss voice are both signals.

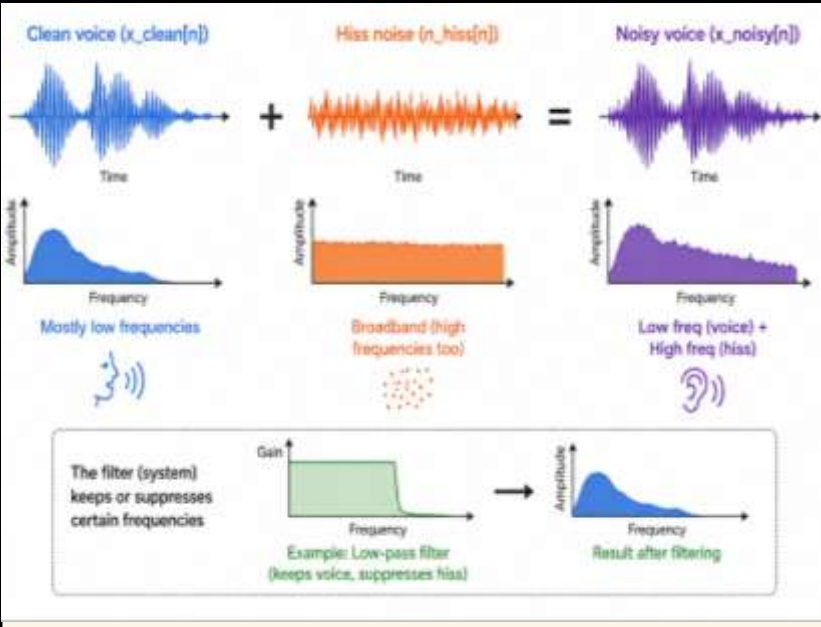


– The noisy voice is a superposition.



– The filter is an input-output system.

Frequency response illustrates what the system keeps or suppresses.



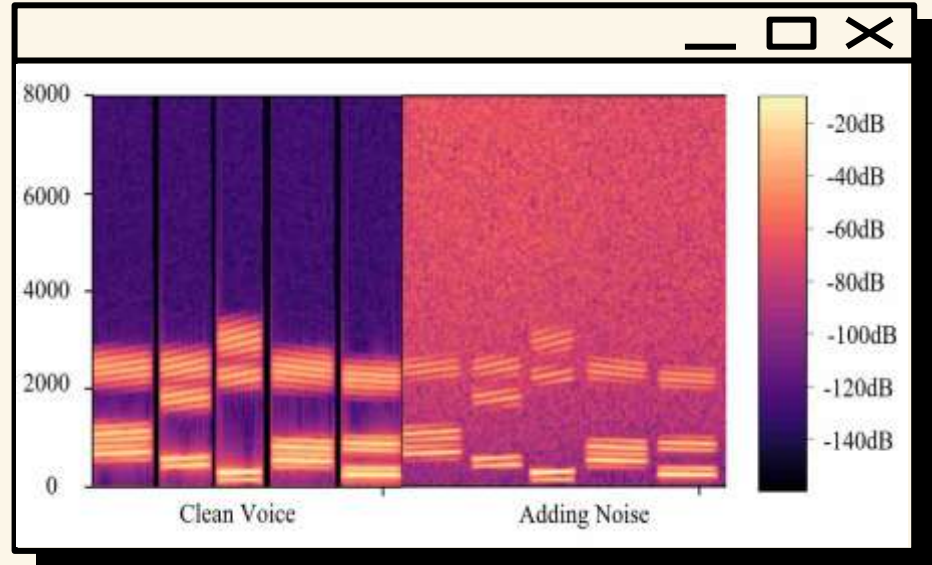
Frequency-Domain Observation

Clean Voice

**Low
Frequency**

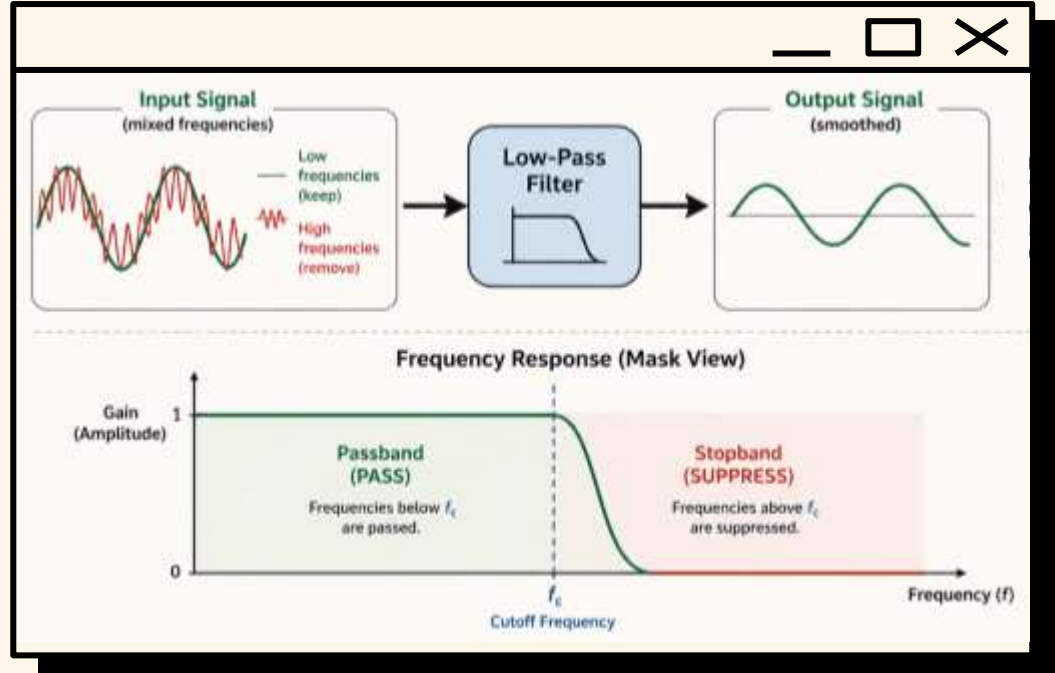
Hiss Noise

**High
Frequency**



Low-Pass Filter

Low-pass filter keeps lower frequencies and attenuates higher frequencies.

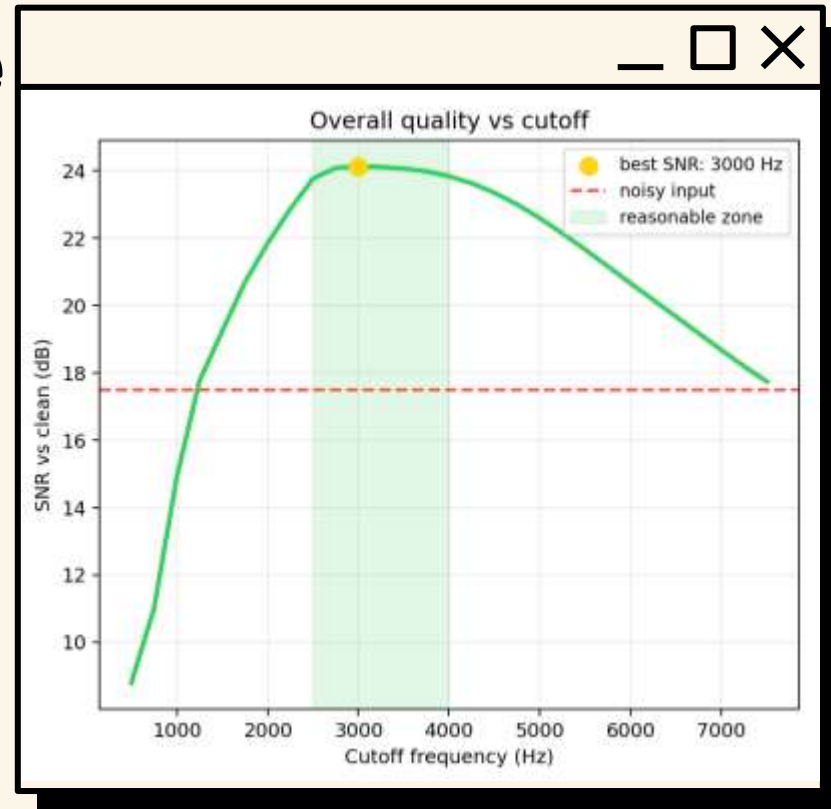


Parameter Choice

Lower cutoff: stronger noise reduction, but speech suffers more.

Higher cutoff: more speech detail, but more hiss remains.

Trade-off !

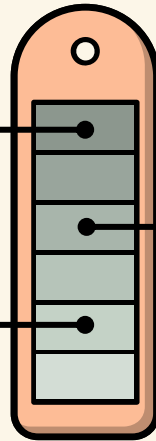


Shortage of low-pass filter

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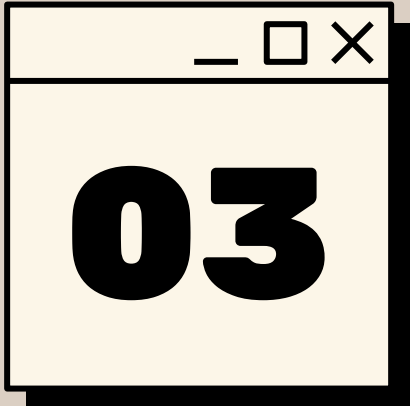
Low-pass uses the same attenuation for the entire signal.

This can reduce hiss, but it can also remove useful speech detail.

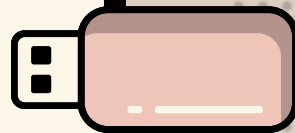


It does not know the signal is speech-dominant or noise-dominant.

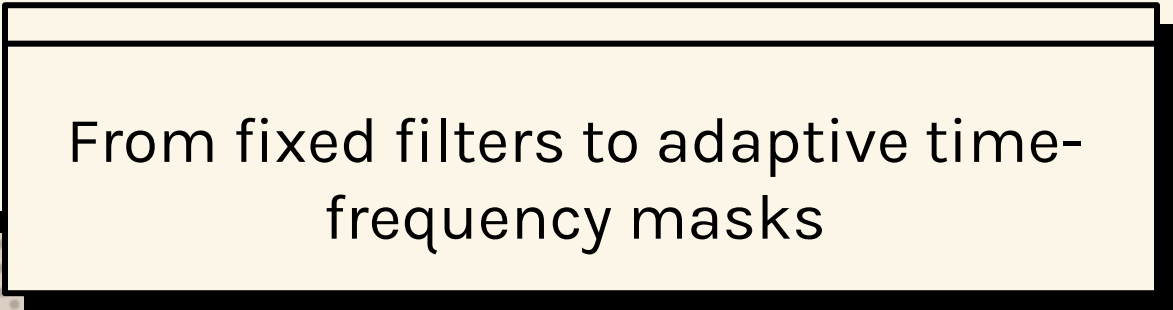
This motivates a more flexible method.



03



From Fixed to Adaptive



From fixed filters to adaptive time-frequency masks



~~~~ Frequency-Domain Masking >>>>



$$Y(f) = M(f)X(f)$$

Specified gains to each frequency bin.



$$Y(t,f) = M(t,f)X(t,f)$$

The mask changes with both time and frequency.



$$M \approx 1$$

Dominated speech means little attenuation.

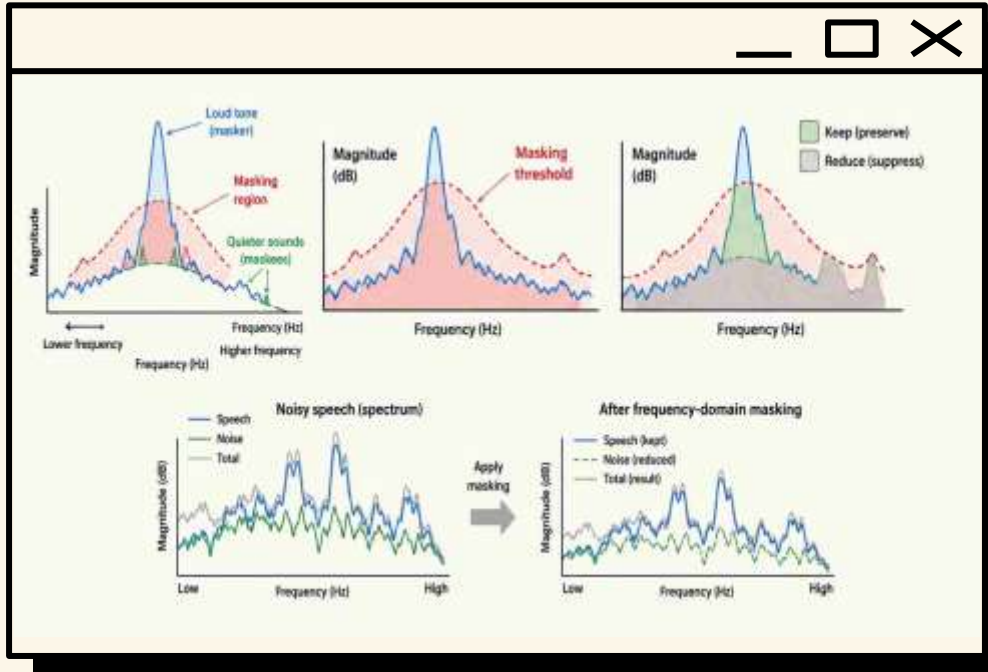


$$M \approx 0$$

Dominated noise means strong reduction.

Frequency-domain mask

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Frequency-domain masking makes local decisions and can better balance noise reduction with speech clarity.

Wiener-Style Mask

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$$M(f) = \frac{P_{\text{speech}}(f)}{P_{\text{speech}}(f) + P_{\text{noise}}(f)}$$

## Adaptive Behavior

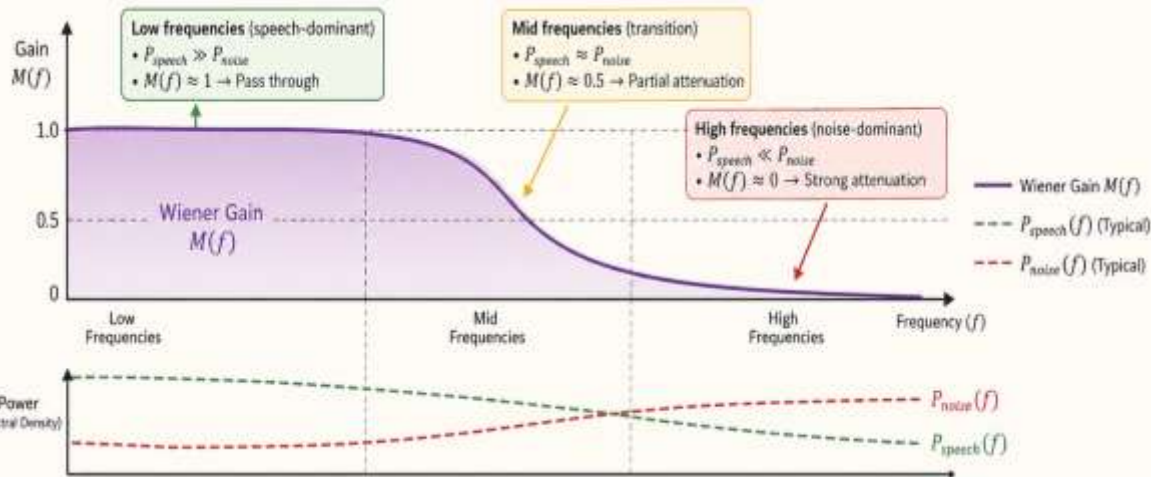
Different frequencies receive different gain values instead of using one fixed cutoff.

## Practical Benefits

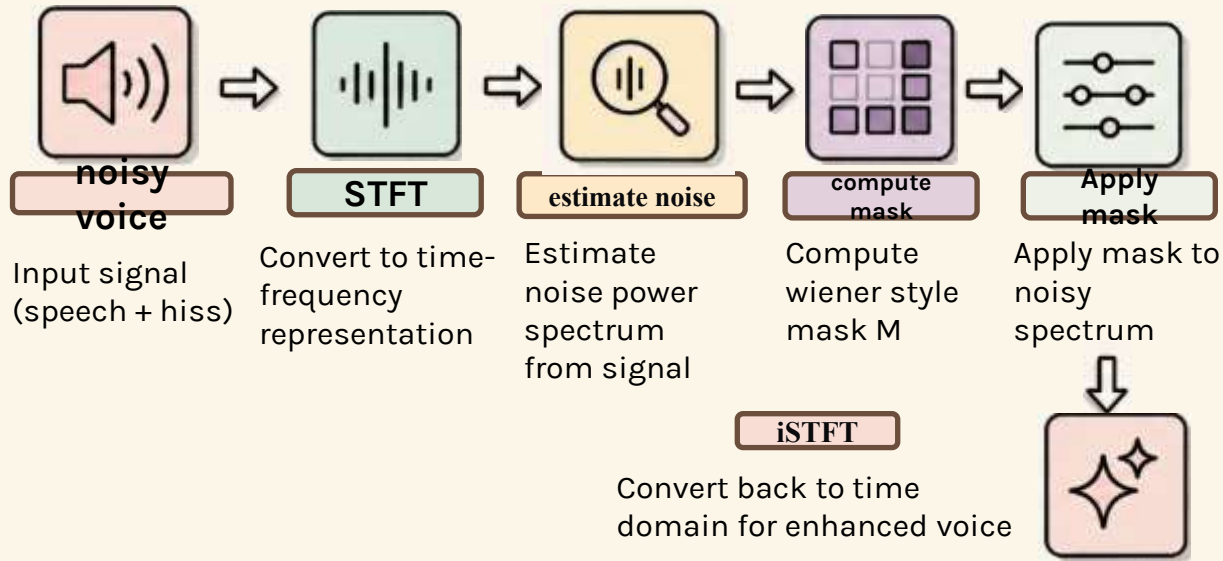
This often gives better clarity than simple low-pass filtering while still reducing hiss noise



# Wiener gain



# Improved method workflow



From fixed filters to adaptive time-frequency masks

# Demo



## 1. Clean voice



Reference of the original speech

## 2. Noisy voice



Speech mixed with  
hiss noise

## 3. Low-pass result



After low-pass filtering:  
noise reduced, but some  
speech details lost

## 4. Mask result



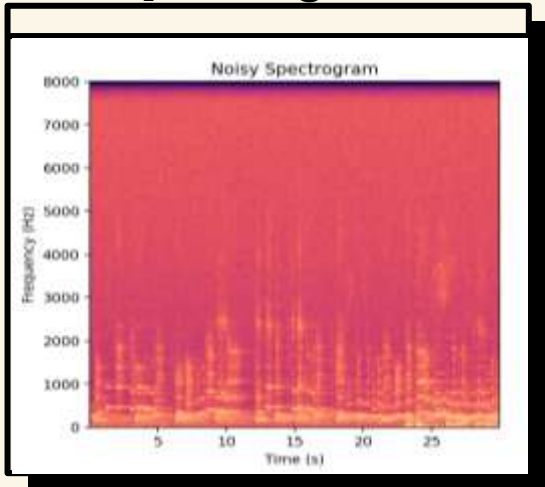
After mask filtering:  
noise suppressed while  
speech details preserved

# Result comparison

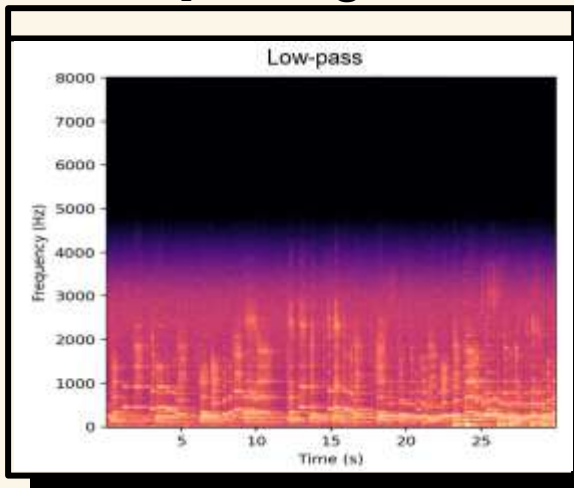
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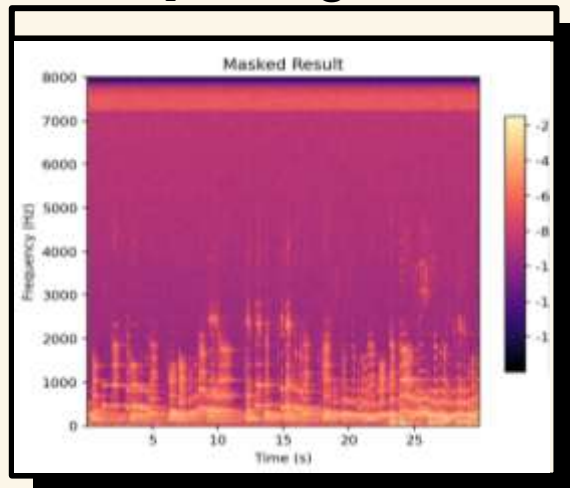
Noisy voice spectrogram



Low-pass result spectrogram



Mask result spectrogram



Conclusions

A simple filter can become a richer



01

Signals and Systems Perspective

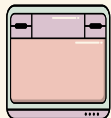
Modeling the voice denoising as a Signals and Systems problem.



02

Baseline: Low-pass Filtering

Low-pass filtering provides a simple baseline by removing high-frequency noise.



03

Frequency-Domain Masking

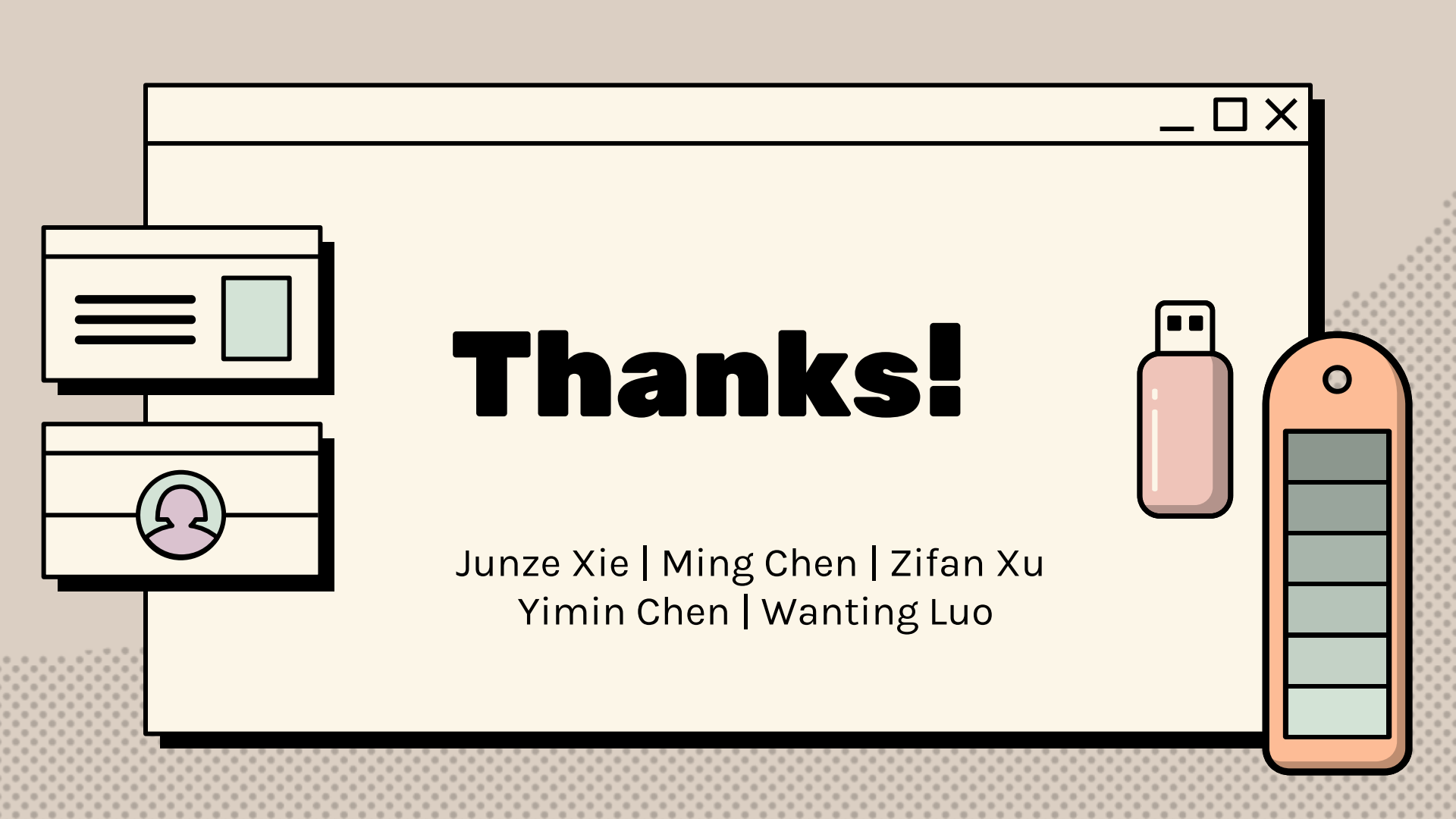
Allowing different gains for different frequencies.



04

Wiener-Style Mask

Designing frequency-dependent attenuation based on the power of speech and noise.



Thanks!

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